• Live on Base Mainnet • $2.20B+ Monitored Debt

The High-Frequency Risk & OEV Infrastructure for Modular DeFi

Sub-20ms parametric stress-testing and liquidation surveillance protecting MetaMorpho vault curators, credit desks, and institutional capital.

$2.20B	~15 ms	94.38%	100.0%
MONITORED DEBT (BASE)	DECISION COMPUTE LATENCY	VERIFIED SVR RECAPTURE	30-DAY VERIFIED UPTIME

Founders: Volkan Dağlı (CTO) • Dr. Zerrin Dağlı (Advisor) • Dağhan Dağlı

Executive Line: ask@answerr.me • radar.werracle.com

Modular Credit Unbundled Without Real-Time Risk Rails

Protocols like Morpho Blue and Aave v3 have unlocked billions in modular lending, but created massive risk blindspots.

⚡ 1. Fragmentation & OEV Leakage

Instead of monolithic pools, lending is scattered across hundreds of isolated markets. During market drops, liquidation bonuses leak to centralized builder cartels via gas wars. On Ethereum L1, **95.9% of builder payments** concentrate in extreme tail liquidations.

📉 2. Curators Flying Blind

MetaMorpho curators (managing $1.5B+ TVL) allocate capital across multiple markets. Current analytics rely on static daily snapshots that cannot simulate real-time collateral drops, cascading liquidations, or pro-rata bad debt before market shocks hit.

🔒 3. Institutional Capital Demands Proof

Institutional credit desks and enterprise treasuries require continuous parametric stress-testing, verifiable on-chain audits, and sub-second loss-given-default (LGD) metrics before deploying capital into permissionless DeFi.

Werracle Reflex Decision Engine: Real-Time Risk DaaS

A production-hardened pipeline connecting Layer-2 flashblocks directly to institutional risk models.

01. L2 Flashblock Surveillance

Direct ingestion of Base 200ms flashblocks and Morpho Blue GraphQL events. Tracks 68,700+ open positions and isolates 45 unlisted/synthetic markets to deliver pure, unpolluted data.

02. Parametric Shock Simulation

Applies real-time collateral price shocks (0%, -5%, -10%, -15%, -20%, -30%) across every live borrower. Evaluates exact solvency against LLTV and Liquidation Incentive Factors (LIF).

03. Verified SVR OEV Recapture

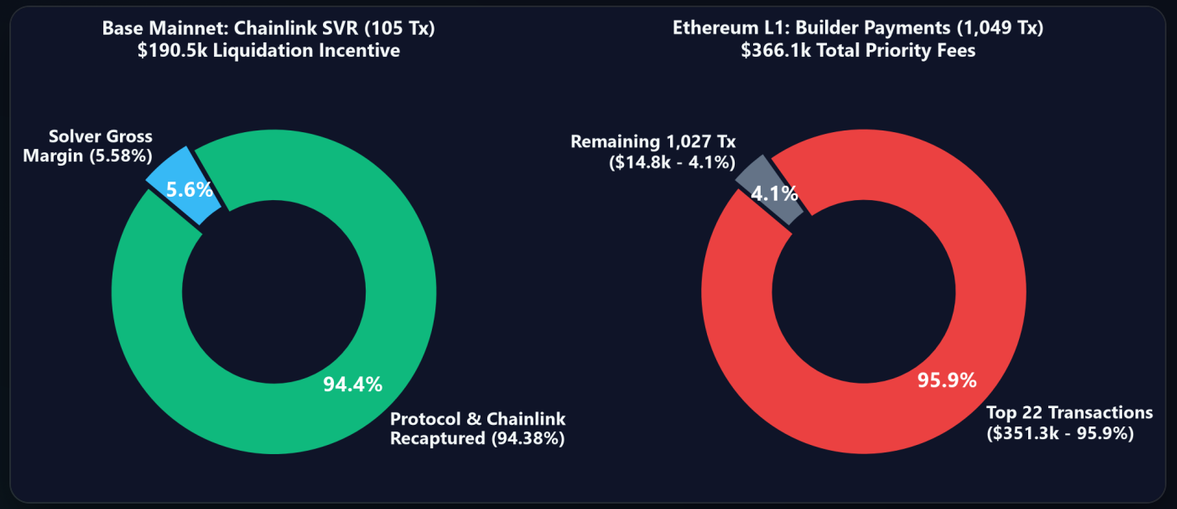
Calibrates expected liquidation recoveries against our empirical baseline proving **94.38% value recapture** via Chainlink Smart Value Recapture (SVR) proceeds contracts.

Live Curator API: `GET https://api.werracle.com/v1/radar/curator/steakhouse`

HTTP 200 OK • Live Tested

Groundbreaking Empirical Validation: 94.38% Recapture

Published research monograph auditing 105 verified SVR liquidation transactions ($1.90M debt repaid) on Base.



94.38% ($179.8k)

OEV Pool Recaptured into Protocol Proceeds

Median transaction recaptures **98.30%** of the liquidation bonus.

+1.26% Median

Solver Gross Margin at Oracle Prices

Fierce competition prevents solver cartels (Top-3 = 77.05%, HHI = 2,216).

95.94% L1 Fat-Tail

22 Largest L1 Liquidations Account for $351.3k in Fees

Empirically proves that Layer-2 SVR eliminates the L1 MEV priority fee black hole.

Active Market Surveillance on Base Mainnet

Zero mock data. Full automated synchronization from Morpho Blue GraphQL and Base RPC nodes.

$2,197,760,378	38 Markets	68,727	45 Markets
CLEAN MONITORED DEBT	WHITELISTED & LISTED	OPEN BORROWER POSITIONS	UNLISTED / SPAM FILTERED

- **Enterprise Data Hygiene**

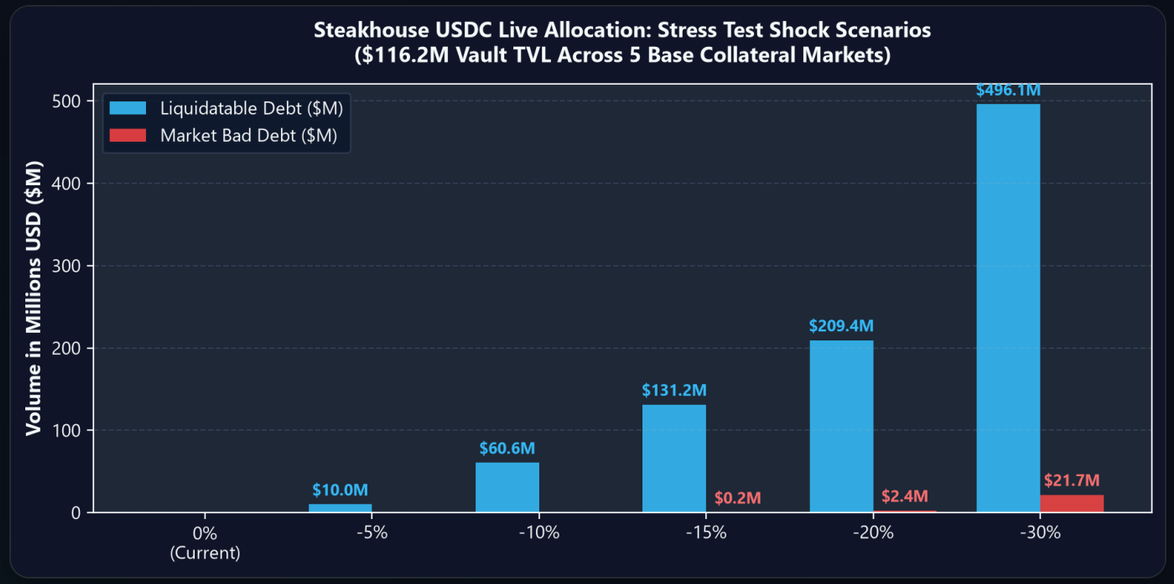
Raw Morpho queries contain $2.31B in synthetic, unlisted markets (e.g. HERMES, test_cbBTC). Werracle is the **only engine** that filters unlisted and unrecognized collateral, providing institutions with pure, audited credit figures.

- **Independent Frankfurt Health Node**

A dedicated remote monitoring daemon in Frankfurt (`mechsrv`) pings the API every 5 minutes and maintains hourly hot-replica database backups. 100% 30-day uptime and 191ms median global latency.

Institutional Stress-Testing for MetaMorpho Curators

Simulating real-time loss-given-default (LGD) across live Steakhouse, Gauntlet, and Re7 vaults.



Case Study: Steakhouse USDC ($116.2M TVL)

Directly ingests live allocations across 5 collateral markets (cbBTC, cbETH, wstETH, etc.) and models collateral price drops.

$0.00
CURRENT BAD DEBT

1.06%
WORST-CASE TVL LOSS (-30%)

Pro-Rata Attribution: Under a -20% collateral shock, $209.4M in debt becomes liquidatable, causing $2.41M in market-wide bad debt. Steakhouse's pro-rata loss remains near $0.00 due to senior allocation tiers.

Recurring Enterprise Tiers & High-Margin Unit Economics

Zero-friction developer onboarding combined with high-value institutional curation contracts.

Free / Public Goods

$0

- Public status & overview API
- Supported vault registry
- Zenodo & HF open data (CC-BY)
- Funnel for institutional leads

Pro Researcher

$149 / mo

- Full REST API access
- 600 requests / minute
- Historical SVR & liquidation logs
- Searchers & MEV researchers

Institutional Curator

$499 / mo

- Dynamic vault stress-testing API
- 3,000 requests / minute
- Parametric shock simulations
- 99.9% uptime SLA & priority support

- **Dual Compliant Billing Rails**

Paddle (Merchant of Record) for automated global cards & VAT + ITouch SWIFT bank invoices for institutional enterprise + EIP-191/1271 signed Base USDC.

- **Tax Advantage: 100% Export Exemption**

Headquartered in Çukurova Teknokent: Software exports to foreign institutions qualify for **100% Corporate Tax exemption** (Pres. Decree 11257).

A Multi-Billion Dollar On-Chain Credit Expansion

As institutional RWAs and decentralized credit migrate to Layer-2s, real-time risk infrastructure becomes mandatory.

TAM • 2028 HORIZON

$45B

Total institutional on-chain lending, credit delegation, and RWA structured debt volume.

SAM • ACTIVE MODULAR TVL

$8.5B

Active TVL across modular and isolated lending protocols (Morpho Blue, Aave v3, Fluid, Euler v2).

SOM • ANNUAL RISK SPEND

$120M

Annual institutional spend on specialized risk curation, liquidation telemetry, and automated stress testing.

Key Industry Tailwind: Morpho Blue's Hyper-Growth

Morpho Blue has expanded from $0 to over $1.5B+ in TVL in under 18 months, becoming the collateral backend for Coinbase, Spark, and Steakhouse. Every new isolated market deployed creates another customer touchpoint for Werracle Radar.

Positioned at the Intersection of High-Frequency & Risk

Legacy risk providers offer slow, expensive consulting. Werracle offers sub-second, automated DaaS APIs.

Key Metric / Capability	Werracle Radar	Chaos Labs / Gauntlet	Block Analitica	Dune Analytics
Latency / Resolution	~15ms Decision Compute	Periodic / Block-level	Daily Static Reports	Seconds to Minutes
Live Dynamic Vault Shocks	Parametric (0 to -30%)	Custom Internal Models	Static Spreadsheets	No (Historical Only)
Verified SVR / OEV Recapture	94.38% Calibrated	None	None	Basic Dashboards
Pricing & Accessibility	$149 – $499 / mo (Self-Serve)	$50k – $250k / yr (Consulting)	$30k+ / yr (Enterprise)	$4,800 / yr (General SQL)
IP Protection	TR 2026/016285 Patent & BSL	Proprietary	Proprietary	Open SQL / Queries

Patent & Licensing Moat

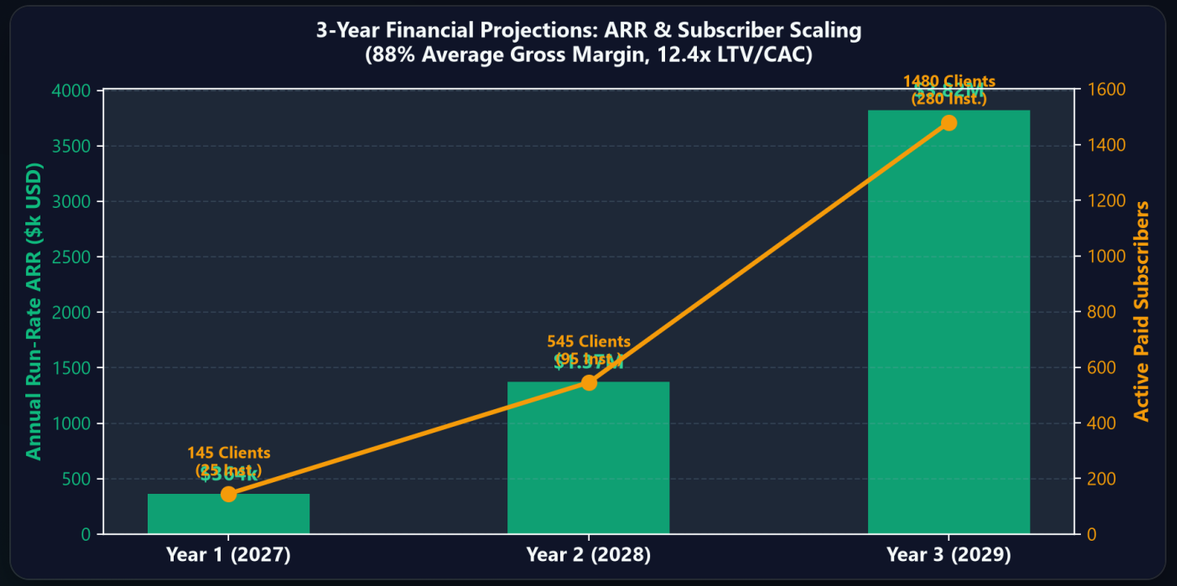
Core procedural execution engine is protected under pending patent application **TÜRKPATENT TR 2026/016285** and licensed under Business Source License 1.1 (BSL 1.1 until 2030-01-01, converting to Apache 2.0).

Data Monopoly in L2 OEV

First-mover advantage as the publisher of the canonical empirical SVR dataset. Curators and protocols build their baseline risk assumptions directly on our open Zenodo decoders.

Predictable, High–Margin Software Expansion

Minimal infrastructure overhead drives an 88% gross margin with rapid compounding ARR.



Year 1 (2027) Run–Rate — **$364,260 ARR**

120 Pro Subscribers • 25 MetaMorpho Curators & Risk Desks • Breakeven achieved in Month 8.

Year 2 (2028) Run–Rate — **$1,373,460 ARR**

450 Pro Subscribers • 95 Institutional Clients across Base, Arbitrum, and Ethereum L1.

Year 3 (2029) Run–Rate — **$3,822,240 ARR**

1,200 Pro Users • 280 Institutional Curators • Standard risk layer for institutional DeFi.

Engineered by Formal Verification & Quantitative Researchers

Bridging theoretical econometrics, high-frequency blockchain engineering, and enterprise delivery.

Volkan Dağlı

Co-Founder & CTO / Lead Developer

Co-Founder & CTO at ITOUCH BİLİŞİM SİSTEMLERİ LTD. ŞTİ. Formal verification researcher affiliated with Anadolu University. Lead architect of Werracle Reflex Engine and author of 4 verified Zenodo open-science monographs.

Dr. Zerrin Dağlı

Academic & Econometric Advisor

Academic advisor affiliated with Mersin University and Yusuf Kalkavan Anadolu Lisesi. Quantitative methodology specialist leading the statistical verification and econometric modeling for liquidation dynamics.

Dağhan Dağlı

Student Researcher & Contributor

Toros Science High School, MEV (Toros University). Focus on smart contract unit testing, RPC decoding pipelines, and automated liquidation simulation environments.

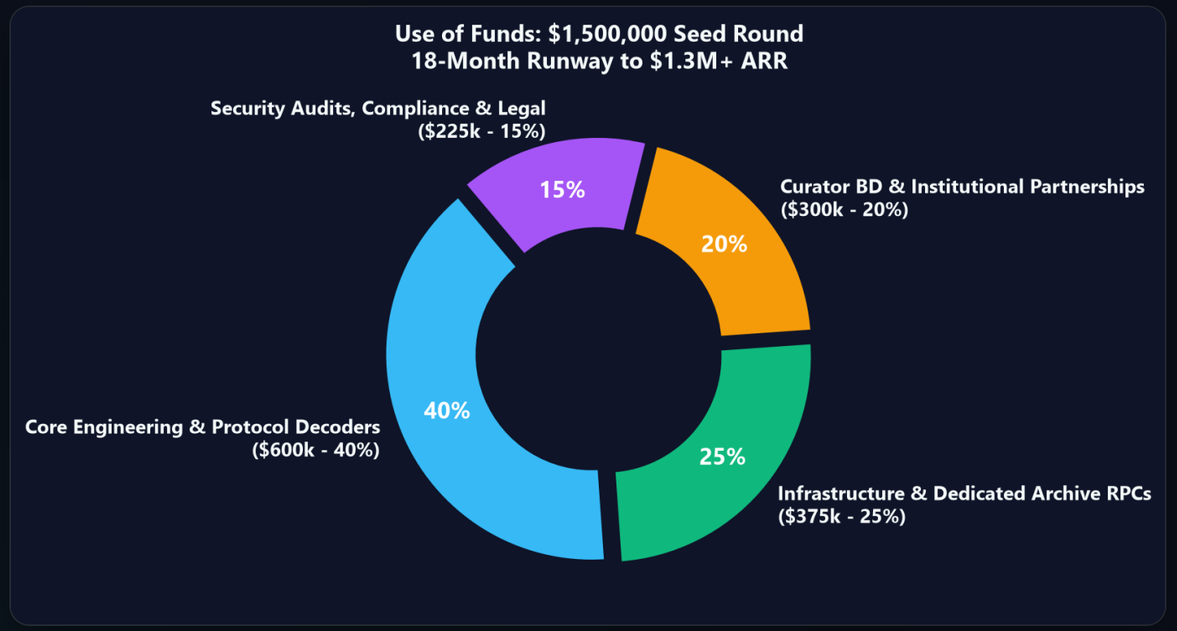
Çukurova Teknokent
Official technology incubator & R&D lab support.

TÜRKPATENT TR 2026/016285
Registered patent application for procedural arbitration.

100% Tax Advantage
Corporate export tax exemption under Pres. Decree 11257.

Raising $1.5M Seed to Scale Institutional DeFi Risk

18–month runway to reach $1.3M+ ARR, multi-chain dominance, and deep curation market penetration.



Engineering Expansion (40% – $600k)

Hire 3 senior Rust/Go EVM systems engineers to scale cross-chain stream processing across Arbitrum, Optimism, and Ethereum L1.

Dedicated Archive Infrastructure (25% – $375k)

Deploy proprietary bare-metal Archive Nodes and sub-second WebSocket distribution clusters for institutional low-latency feeds.

Institutional BD & Curators (20% – $300k)

Direct integrations with MetaMorpho curators (Steakhouse, Gauntlet, Re7), Aave risk DAOs, and institutional lending desks.

Security Audits & Compliance (15% – $225k)

Complete SOC2 Type II compliance, enterprise security hardening, and legal advisory for international institutional contracts.